

Histidine-rich Glycoprotein Binds Fibrin(ogen) with High Affinity and Competes with Thrombin for Binding to the γ' -Chain*

Received for publication, April 25, 2011, and in revised form, June 28, 2011. Published, JBC Papers in Press, July 8, 2011, DOI 10.1074/jbc.M111.253831

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Histidine-rich glycoprotein (HRG) is an abundant protein that binds fibrinogen and other plasma proteins in a Zn^{2+} -dependent fashion but whose function is unclear. HRG has antimicrobial activity, and its incorporation into fibrin clots facilitates bacterial entrapment and killing and promotes inflammation. Although these findings suggest that HRG contributes to innate immunity and inflammation, little is known about the HRG-fibrin(ogen) interaction. By immunoassay, HRG-fibrinogen complexes were detected in Zn^{2+} -supplemented human plasma, a finding consistent with a high affinity interaction. Surface plasmon resonance determinations support this concept and show that in the presence of Zn^{2+} , HRG binds the predominant γ_A/γ_A -fibrinogen and the γ -chain elongated isoform, γ_A/γ' -fibrinogen, with K_d values of 9 nM. Likewise, ^{125}I -labeled HRG binds γ_A/γ_A - or γ_A/γ' -fibrin clots with similar K_d values when Zn^{2+} is present. There are multiple HRG binding sites on fibrin(ogen) because HRG binds immobilized fibrinogen fragment D or E and γ' -peptide, an analog of the COOH terminus of the γ' -chain that mediates the high affinity interaction of thrombin with γ_A/γ' -fibrin. Thrombin competes with HRG for γ' -peptide binding and displaces ^{125}I -HRG from γ_A/γ' -fibrin clots and vice versa. Taken together, these data suggest that (a) HRG circulates in complex with fibrinogen and that the complex persists upon fibrin formation, and (b) by competing with thrombin for γ_A/γ' -fibrin binding, HRG may modulate coagulation. Therefore, the HRG-fibrin interaction may provide a novel link between coagulation, innate immunity, and inflammation.

Fibrinogen is the soluble precursor of fibrin, a critical component of blood clots that endows them with strength and elasticity. Fibrinogen is a glycoprotein composed of three pairs of polypeptide chains, termed $\text{A}\alpha$, $\text{B}\beta$, and γ , that are connected

by disulfide bonds (1). Approximately 10–15% of circulating fibrinogen has a variant γ -chain termed the γ' -chain, which results from differential processing of the γ_A -chain mRNA transcript (2–4). The γ' -chain is distinguished from the γ_A -chain by the presence of an acidic 20-residue extension at its COOH terminus (2, 5).

Thrombin catalyzes the conversion of fibrinogen to insoluble fibrin, and during this process some thrombin remains bound to the fibrin network (6). Exosites 1 and 2 are two regulatory domains that flank the active site of thrombin and mediate its binding to fibrin (7, 8). Exosite 1 of thrombin interacts with the central E-domain of fibrin, whereas exosite 2 binds only to the COOH terminus of the γ' -chain. Consequently, thrombin binds γ_A/γ_A -fibrin in a univalent fashion with a K_d value of 2–4 μM . In contrast, both exosites are engaged when thrombin binds to γ_A/γ' -fibrin, resulting in a higher affinity interaction (K_d value of 0.08–0.18 μM) (5, 9). Fibrin-bound thrombin remains active, and the protease is protected from inhibition by fluid-phase inhibitors, such as antithrombin and heparin cofactor II (6). Because of its bivalent interaction with γ_A/γ' -fibrin, thrombin bound to γ_A/γ' -fibrin is more protected from inhibition by fluid-phase inhibitors than thrombin bound to γ_A/γ_A -fibrin (10).

Like thrombin, histidine-rich glycoprotein (HRG)³ binds to fibrinogen and is incorporated into fibrin clots (11). Although the plasma concentration of HRG ranges from 1.6 to 2 μM , the concentration in platelet-rich thrombi may be higher because HRG is stored in the alpha granules of platelets and is released when platelets are activated (12, 13). A 75-kDa glycoprotein, HRG, is composed of two NH_2 -terminal cystatin-like domains, a central histidine-rich region (HRR) flanked by two proline-rich regions, and a COOH-terminal domain (14). In addition to fibrinogen, HRG also binds plasminogen, heparan sulfate, and divalent cations, such as Zn^{2+} (12, 15, 16). Therefore, HRG is hypothesized to be an important accessory or adapter protein that brings different ligands together under specific conditions (14).

HRG-deficient mice exhibit a shorter prothrombin time and accelerated fibrinolysis compared with wild-type mice, raising

* This work was supported in part by the Canadian Institutes of Health Research (MOP 3992, MOP 102735, and CTP 79846), the Heart and Stroke Foundation of Ontario (T4792 and T4730), the Heart and Stroke Foundation of Canada (a Novel and Exploratory Research Fund grant), and the Ontario Research and Development Challenge Fund.

¹ Supported by a Heart and Stroke Foundation of Ontario Graduate Student Award.

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³ The abbreviations used are: HRG, histidine-rich glycoprotein; HRR, histidine-rich region; Fab, fragment antibody binding; FPRck, D-Phe-Pro-Arg chloromethyl ketone; YPRck, D-Tyr-Pro-Arg chloromethyl ketone; HBS, HEPES-buffered saline; γ' -peptide, synthetic analog of the COOH-terminal portion of the γ' -chain of γ_A/γ' -fibrinogen; RU, response units; Req, amount of ligand bound to immobilized protein at equilibrium; SPR, surface plasmon resonance.

the possibility that HRG modulates coagulation and fibrinolysis (17). In addition to its potential role in hemostasis, HRG also has been implicated in innate immunity and inflammation (18). HRG exhibits antifungal and antimicrobial activity *in vitro*, and these activities are enhanced at low pH or in the presence of Zn^{2+} , conditions that promote ligand binding (19, 20). The antimicrobial activity of HRG has also been demonstrated *in vivo* and appears to be fibrin-dependent. Thus, compared with wild-type mice, HRG-deficient mice are more susceptible to the lethal effect of *Streptococcus pyogenes* infection and are rescued with HRG supplementation (21). This phenomenon is fibrin-dependent because fibrin is essential for HRG-mediated bacterial entrapment and killing, processes that prevent bacterial dissemination. In addition, the HRG-fibrin interaction modulates inflammation because HRG-deficient mice exhibit attenuated abscess formation in response to subcutaneous injection of bacteria. Based on these findings, it has been postulated that HRG plays a fibrin-dependent role in both inflammation and innate immunity (21).

Despite emerging evidence that the HRG-fibrin(ogen) interaction is physiologically important, little is known about the biochemical foundation of this interaction or its functional consequences. To address these gaps in knowledge, we set out to (a) quantify the binding of HRG to fibrin(ogen), (b) identify the HRG binding domains on fibrin(ogen), and (c) determine whether there is overlap between the HRG and thrombin binding domains on fibrin(ogen).

EXPERIMENTAL PROCEDURES

Materials

Reagents—Human thrombin, prothrombin, and plasminogen-free fibrinogen were from Enzyme Research Laboratories (South Bend, IN). Plasmin and factor XIII were from Haematologic Technologies Inc. (Essex Junction, VT). Prionex was from Pentapharm (Basel, Switzerland). A 20-amino acid analog of the COOH terminus of the γ' -chain of fibrinogen, γ' -peptide (VRPEHPAETEDSLYPEDDL), was prepared by Bachem Bioscience, Inc. (King of Prussia, PA), and a sheep antibody against this peptide was from Affinity Biologicals (Ancaster, ON). The two Tyr residues within the γ' -peptide were modified with phosphate groups in place of sulfate to enhance stability (22). The γ' -peptide-directed IgG was subjected to affinity chromatography using immobilized γ' -peptide (SulfoLink Immobilization Kit for Peptides, Thermo Scientific, Rockford, IL). D-Phe-Pro-Arg chloromethyl ketone and D-Tyr-Pro-Arg chloromethyl ketone (FPRck and YPRck, respectively) were from EMD Chemicals (Gibbstown, NJ). Fibrinogen was rendered factor XIII-free, and γ_A/γ_A - and γ_A/γ' -fibrinogen were separated by fractionation on DEAE-Sepharose (GE Healthcare) and characterized as previously described (10, 23). HRG was purified by metal-chelate chromatography, and HRG-deficient plasma was prepared as described previously (24). Non-immune sheep IgG and a human HRG-directed IgG from sheep were prepared by Affinity Biologicals, and the HRG-specific IgG fraction was isolated by affinity chromatography using immobilized HRG (24). Unless otherwise specified, other reagents were from Sigma.

Preparation of Fibrinogen Fragments—Fragment X was prepared by limited plasmin digestion of fibrinogen (23). Fragments D and E were generated by plasmin digestion of γ_A/γ_A -fibrinogen (25). Digested material was applied to a 12-ml UNO Q-12 ion exchange column (Bio-Rad) using a Bio-Rad Biologic Duoflow system at a flow rate of 5 ml/min. To elute non-specifically bound proteins, the column was washed with 30 ml of 0.02 M sodium phosphate, 0.01 M citric acid, pH 7.6. Fragment D was eluted with 100 ml of 0.1 M sodium phosphate, 0.05 M citric acid, pH 5.0, whereas fragment E was subsequently eluted with 40 ml of the same buffer at pH 4.4 (26). Protein-containing fractions were identified by absorbance at 280 nm and pooled. Purified fragments were dialyzed into 10 mM Hepes-NaOH, 150 mM NaCl, pH 7.4 (HBS) and concentrated. Final concentrations were determined at 280 nm (27). The integrity of the fragments was assessed by SDS-PAGE analysis on 4–15% polyacrylamide gels (Ready-Gel, Bio-Rad) under reducing and non-reducing conditions. Samples were stored in aliquots at -80°C .

Preparation of γ' -directed IgG Fab Fragments—For some experiments, fragment antibody binding (Fab) regions from the affinity-purified γ' -peptide-directed IgG were generated by papain digestion (28), isolated with a Fab preparation kit (Pierce), assessed for purity by SDS-PAGE analysis, and then concentrated and dialyzed against HBS.

Labeling of Proteins— γ' -Peptide was labeled with fluorescein isothiocyanate as previously described (9). α -Thrombin was radiolabeled by reaction with ^{125}I -labeled YPRck, and HRG was radiolabeled with Na^{125}I (McMaster University Nuclear Reactor, Hamilton, ON) using Iodo-beads (Pierce) as described (29). ^{125}I -YPRck-thrombin and ^{125}I -HRG concentrations were 5–6 μM , as determined by absorbance at 280 nm, with radioactivity of 500,000–800,000 cpm/ μg of protein. FPRck-thrombin was prepared as described (30).

Methods

Surface Plasmon Resonance (SPR)—The interaction of HRG with immobilized γ_A/γ_A - or γ_A/γ' -fibrin(ogen), biotinylated- γ' -peptide, or fragments X, D, or E was assessed by SPR using a BIAcore 1000 (GE Healthcare) as previously described (24, 30) but with some modifications. Briefly, proteins were covalently linked to separate flow cells of a carboxymethylated dextran (CM4) biosensor chip at a flow rate of 5 $\mu\text{l}/\text{min}$ using an amine coupling kit (GE Healthcare). Proteins were immobilized using 10 mM acetate buffer at varying pH values to maximize adsorption. γ_A/γ_A - or γ_A/γ' -fibrinogen or fragments X and D were immobilized at pH 5.5 to ~ 3000 –7000 response units (RU). Fragment E was immobilized at pH 4.5 to ~ 2000 –3000 RU. For fibrin binding studies, immobilized γ_A/γ_A - or γ_A/γ' -fibrinogen was converted to fibrin by three successive 60-min injections of 100 nM thrombin at 5 $\mu\text{l}/\text{min}$ (30). To prepare streptavidin-conjugated CM4 flow cells, 0.4 mg/ml streptavidin (Sigma) at pH 4.5 was injected. Biotinylated γ' -peptide (30) was adsorbed to the immobilized streptavidin to 200–300 RU. Remaining reactive groups were neutralized with 1 M ethylenediamine, and non-specifically adsorbed proteins were removed by treatment with 0.5 M NaCl. An unmodified flow cell served as the control. All SPR procedures were done in HBS containing 0.005%

Tween 20 and 2 mM CaCl₂, and flow cells were regenerated with 250 mM imidazole and 2 mM EDTA between runs.

To measure the affinity of HRG for immobilized fibrinogen, fibrin, or fibrinogen fragments, aliquots of HRG (0–1 μM) in buffer containing 20 μM ZnCl₂ were injected at a flow rate of 30 μl/min. To quantify Zn²⁺ dependence, binding of 200 nM HRG to immobilized γ_A/γ_A- or γ_A/γ'-fibrinogen was monitored in the presence of varying concentrations of ZnCl₂ (0–60 μM) using dual injection mode. The binding of thrombin to immobilized fragment D or E was monitored by injection of FPRck-thrombin (0–15 μM) into flow cells.

The γ'-peptide-directed IgG was used to assess the contribution of the γ'-chain to the interaction of HRG with immobilized γ'-peptide or fibrinogen. A saturating amount of γ'-peptide-directed IgG or a non-immune IgG (2 μM) was first injected into flow cells containing immobilized γ'-peptide or γ_A/γ'- or γ_A/γ_A-fibrinogen. The binding of HRG to fibrinogen or fibrinogen fragments was then measured as described above, except that flow cells were re-saturated with 0.5 μM γ'-peptide-directed IgG or control IgG before each HRG injection.

Binding of HRG to immobilized γ'-peptide in the absence or presence of FPRck-thrombin was monitored using a BIAcore T200. Biotinylated γ'-peptide was adsorbed to streptavidin-conjugated CM4 flow cells to 100 RU in the presence of 20 μM ZnCl₂ at a flow rate of 5 μl/min. An unmodified flow cell served as a control. Using dual injection mode, 1 μM HRG was injected at 10 μl/min followed by a second injection of FPRck-thrombin or prothrombin at concentrations ranging from 0 to 8 μM. Flow cells were regenerated with 250 mM imidazole, 2 mM EDTA, and 1 M NaCl. All experiments were performed at least twice.

SPR Data Analysis—*K_d* values were determined by kinetic analysis of on- and off-rates of HRG binding to immobilized ligands using Scrubber2 version 2.0a (Bio-Logic Software Co., Campbell, Australia) as described previously (24, 30). For further assessment of binding, the amount of HRG bound at the equilibrium position (Req) was determined using the Langmuir 1:1 binding model (BIAevaluation software Version 3.2) and was plotted against the titrant concentration. Molar stoichiometries were determined as described in the BIAtechnology handbook (BIAcore 1000). The correction factor to account for the orientation of the immobilized fibrinogen and fibrin was 0.25, which corresponds to 25% of the amount of immobilized fibrin accessible to the γ'-peptide-directed IgG as determined in a separate study (10). The correction factor for immobilized γ'-peptide was 0.7, which corresponds to 70% correct orientation of peptide accessible to an analyte (BIAcore).

Interaction of ¹²⁵I-HRG with Fibrin Clots—In a series of microcentrifuge tubes, γ_A/γ_A- or γ_A/γ'-fibrinogen, in concentrations ranging from 0 to 1.25 μM, was clotted with 10 nM thrombin in the presence of 40 nM ¹²⁵I-HRG as previously described (30). Clots were formed in 20 mM Tris-HCl and 150 mM NaCl, pH 7.4 (TBS), containing 0.005% Tween (TBS-Tween) and 2 mM CaCl₂ and 20 μM ZnCl₂ or 10 μM sodium diethyldithiocarbamate trihydrate. The concentration of ¹²⁵I-HRG bound to fibrin was calculated by subtracting the concentration in the clot supernatant from the value obtained in controls prepared in the absence of fibrinogen. Plots of bound ¹²⁵I-HRG versus fibrin concentration were analyzed by nonlin-

ear regression of a rectangular hyperbola to determine *K_d*. Experiments were performed twice in duplicate.

Effect of Competitors on the Binding of ¹²⁵I-HRG to Fibrin Clots—Clots were formed in TBS-Tween containing 2 mM CaCl₂ and 20 μM ZnCl₂. Fab fragments derived from γ'-peptide-directed IgG were used to assess the contribution of the γ'-chain to ¹²⁵I-HRG binding to clots. After preincubation of 0.25 μM γ_A/γ_A- or γ_A/γ'-fibrinogen with γ'-peptide-directed Fab fragments or control sheep IgG (0–4 μM) for 1 h at 23 °C, 60 nM ¹²⁵I-HRG was added, and clots were generated with 20 nM thrombin. To determine whether HRG and thrombin share fibrin binding sites, the effect of FPRck-thrombin on the binding of ¹²⁵I-HRG was assessed. Samples containing 2 μM γ_A/γ_A- or γ_A/γ'-fibrinogen, 20 nM ¹²⁵I-HRG, and FPRck-thrombin (0–8 μM) in 0.25% Prionex were clotted with 20 nM thrombin. The effect of varying concentrations of HRG on the binding of ¹²⁵I-YPRck-thrombin was assessed in a reciprocal experiment. Samples containing 2 μM γ_A/γ_A- or 0.25 μM γ_A/γ'-fibrinogen, 20 nM ¹²⁵I-YPRck-thrombin, and HRG (0–2 μM) were clotted with 10 nM thrombin. The fraction of ¹²⁵I-HRG or ¹²⁵I-YPRck-thrombin bound was determined as described above. All experiments were performed twice in duplicate.

HRG Diffusion from Preformed Fibrin Clots—The rate of ¹²⁵I-HRG dissociation from γ_A/γ_A- or γ_A/γ'-fibrin clots was determined as previously described (10, 30). Briefly, fibrin clots were formed around plastic inoculation loops (Bac-Loop, Thermo-Fisher Scientific, Waltham, MA) by adding 10 nM thrombin to solutions containing 5 μM γ_A/γ_A- or γ_A/γ'-fibrinogen, 20 nM factor XIII, and 50 nM ¹²⁵I-HRG in TBS-Tween containing 2 mM CaCl₂ and 20 μM ZnCl₂. After incubation for 45 min at 23 °C, clots were removed and immersed in buffer containing 2 mM CaCl₂ plus 20 μM ZnCl₂, 2 mM CaCl₂ plus 10 μM diethyldithiocarbamate trihydrate, or 2 M NaCl plus 2 mM EDTA. The fraction of clot-associated ¹²⁵I-HRG remaining at various times was determined, and time courses were fit to a two-phase exponential decay curve (Table Curve, Jandel Scientific, San Rafael, CA) (10). Experiments were repeated three times.

Detection of HRG-Fibrinogen Complexes in Plasma—HRG-fibrinogen complexes in plasma were detected using a previously described sandwich ELISA (24) with some modifications. Briefly, 100 μl of fibrinogen-directed capture antibody (Affinity Biologicals) diluted to 20 μg/ml in 50 mM NaHCO₃, pH 9.6, was added to wells of a 96-well Immunolon 4 HBX plate (Thermo Scientific) and incubated overnight at 4 °C. To block nonspecific binding, 200 μl of 10 mg/ml bovine serum albumin (Sigma) was added to each well and incubated for 1 h at 23 °C. Wells were washed 3 times with 150 μl of phosphate-buffered saline (PBS) containing 10 μM ZnCl₂ and 0.1% Tween 20. Normal and HRG-deficient plasma samples were dialyzed against TBS to remove citrate, reconstituted with 18 μM ZnCl₂, and then serially diluted up to 1600-fold with HBS containing 1% ovalbumin, 0.1% Tween 20, 2 mM CaCl₂, 100 nM hirudin (Behring), and 18 μM ZnCl₂. To prepare a reference curve, HRG-fibrinogen complexes were generated by incubating varying concentrations of HRG (0–1.6 μg/ml) with fibrinogen (0–0.8 μg/ml) in a 2:1 molar ratio. 100 μl of each of these mixtures or plasma was added to wells and incubated for 2 h at 23 °C. After three sequential washes with PBS containing ZnCl₂, HRG-fibrinogen

complexes were detected in purified and plasma systems using a HRG-directed IgG-horseradish peroxidase (HRP) conjugate, prepared as specified by the manufacturer using a Lightning-link HRP conjugation kit (Cedarlane, Burlington, ON). After incubation for 1 h at 23 °C, bound HRP conjugates were detected as specified by the supplier. Experiments were repeated three times.

Interaction of HRG with Fluorescein-labeled γ' -Peptide in the Absence or Presence of ZnCl_2 —The binding of 1.1 μM HRG to 0.05 μM fluorescein- γ' -peptide was monitored by fluorescence in the absence or presence of Zn^{2+} using a PerkinElmer Life Sciences LS 50B luminescence spectrometer (9). Briefly, the base-line fluorescence (I_0) was determined at excitation and emission wavelengths (slit widths) of 492 (5 nm) and 532 nm (2.5 nm), respectively, and an emission filter at 515 nm. The mixture was then titrated with aliquots of ZnCl_2 up to 20 μM , and fluorescence intensity (I) was monitored after each addition. I/I_0 values were plotted against the concentration of ZnCl_2 , and the data were subjected to nonlinear regression analysis as previously described (9).

Statistical Analyses—Results are presented as the mean $\pm$ S.D., and the significance of differences in the means was determined using t tests. For these analyses, $p < 0.05$ was considered statistically significant.

RESULTS

Interactions of HRG with γ_A/γ_A - or γ_A/γ' -Fibrinogen—Although HRG has previously been shown to bind fibrinogen (11), the distinction between γ_A/γ_A - and γ_A/γ' -fibrinogen binding has not been investigated. SPR was used to characterize the interaction between HRG and the two isoforms of fibrinogen. γ_A/γ_A - or γ_A/γ' -fibrinogen was immobilized on separate flow cells of a CM4 sensor chip, and an unmodified flow cell served as the control. HRG did not bind either form of fibrinogen in the presence of Ca^{2+} alone (data not shown). Because Zn^{2+} facilitates the binding of ligands to HRG (14), we examined the effect of Zn^{2+} on the HRG-fibrinogen interaction. The Req increased as a function of the Zn^{2+} concentration and saturated at $\sim 30 \mu\text{M}$ ZnCl_2 (Fig. 1). At each Zn^{2+} concentration, more HRG bound to γ_A/γ' -fibrinogen than to γ_A/γ_A -fibrinogen. The apparent K_d values of ZnCl_2 necessary to promote HRG binding to γ_A/γ_A - or γ_A/γ' -fibrinogen were 4.9 ± 0.2 and $1.2 \pm 0.8 \mu\text{M}$, respectively. The Zn^{2+} dependence of the interaction of HRG with fibrin(ogen) is in apparent contradiction to previous work demonstrating HRG binding to fibrin without Zn^{2+} addition (11). Because no binding was detected in the absence of Zn^{2+} in our study, it is likely that the HRG preparation used in the previous report contained sufficient amounts of Zn^{2+} to enable the interaction. For the remainder of the study, ZnCl_2 was used at a concentration of 20 μM . This concentration was chosen because the physiological concentration of Zn^{2+} in plasma ranges from 10 to 20 μM (15, 31). As evidenced from the similarity in saturation profiles illustrated in Fig. 1, more than 80% of HRG is bound to both forms of fibrinogen at 20 μM ZnCl_2 .

To determine the affinity of HRG for fibrinogen, increasing concentrations of HRG were sequentially injected into flow cells containing immobilized fibrinogen in the presence of 20

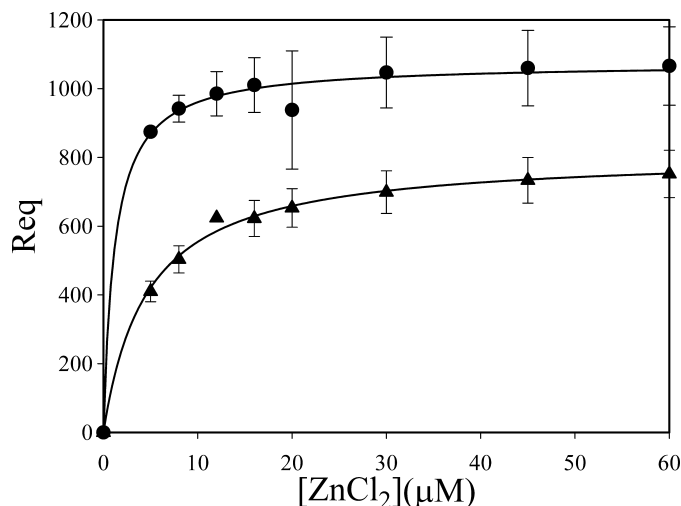


FIGURE 1. Effect of ZnCl_2 on the binding of HRG to γ_A/γ_A - or γ_A/γ' -fibrinogen. γ_A/γ_A -Fibrinogen (triangles) or γ_A/γ' -fibrinogen (circles) was immobilized to 6000–7000 RU on separate flow cells of a CM4 BIAcore chip. An unmodified flow cell served as control. HRG (0.2 μM) was injected into flow cells in the absence or presence of ZnCl_2 at the concentrations indicated. The RU at equilibrium (Req) was calculated and, after background correction, is plotted against the input ZnCl_2 concentrations. Data represent the mean $\pm$ S.D. of two experiments, and lines represent nonlinear regression analyses of the data.

μM ZnCl_2 . The sensograms reveal slow association and dissociation phases for HRG binding to both isoforms of fibrinogen (Fig. 2, A and B). K_d values were obtained by kinetic analysis of the on- and off-rates by globally fitting the binding data. In the presence of Zn^{2+} , HRG binds γ_A/γ_A - and γ_A/γ' -fibrinogen with similar affinity, K_d values of 8.8 ± 0.9 and $8.9 \pm 3.9 \text{ nM}$, respectively (Table 1). These values agree with the previously reported K_d of 6.7 nM determined by immunoassay (11). Plots of calculated Req values for each HRG concentration revealed saturable binding and demonstrated that binding of HRG to γ_A/γ' -fibrinogen was significantly ($p < 0.005$) higher than that for γ_A/γ_A -fibrinogen (Fig. 3), suggesting that 2-fold more HRG is bound to the γ_A/γ' -fibrinogen isoform. The molar stoichiometries for the interaction of HRG with γ_A/γ_A -fibrinogen and γ_A/γ' -fibrinogen are 1.7 ± 0.3 and 3.2 ± 0.5 , respectively (Table 1). Because the extended COOH terminus of the γ' -chain is the feature that distinguishes γ_A/γ' -fibrinogen from γ_A/γ_A -fibrinogen, the increased binding of HRG to γ_A/γ' -fibrinogen suggests that the γ' -chain provides the additional HRG binding site.

Effect of the γ' -Peptide-directed IgG on the Binding of HRG to γ_A/γ_A - or γ_A/γ' -Fibrinogen—To confirm that HRG binds specifically to the γ' -chain of γ_A/γ' -fibrin(ogen), we examined the effect of an affinity-purified IgG directed against this region. As an initial control, we demonstrated specific binding of the antibody to immobilized γ_A/γ' -fibrinogen but not to γ_A/γ_A -fibrinogen (data not shown). We next examined the effect of the antibody on HRG binding to γ_A/γ' - or γ_A/γ_A -fibrinogen. The γ' -chain-directed antibody reduced HRG binding to γ_A/γ' -fibrinogen to that observed with γ_A/γ_A -fibrinogen (Fig. 3). As a control, a sheep non-immune IgG was used; the control IgG had no effect on the binding of HRG to fibrinogen (data not shown). Because the interaction of HRG with γ_A/γ_A -fibrinogen is already of high affinity, the addition of the γ' -peptide-di-

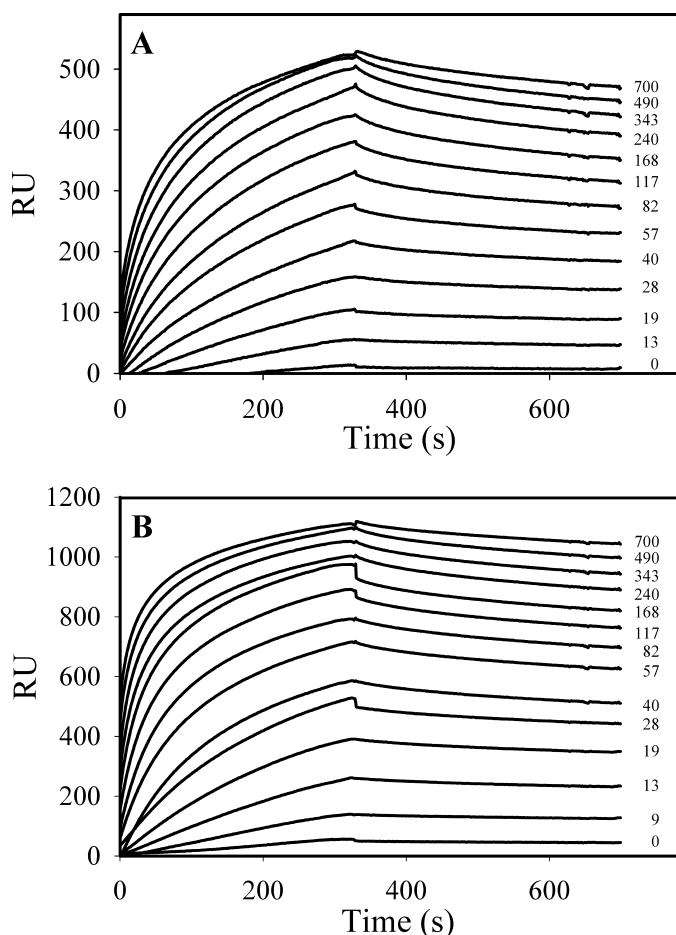


FIGURE 2. Determination of the affinity of HRG for γ_A/γ_A - or γ_A/γ' -fibrinogen in the presence of ZnCl_2 . γ_A/γ_A -Fibrinogen (A) or γ_A/γ' -fibrinogen (B) was adsorbed to separate flow cells on a CM4 chip. HRG (0–700 nM) was injected into flow cells for 300 s in the presence of 20 μM ZnCl_2 , and the cells were then washed with HBS buffer containing 2 mM CaCl_2 and 20 μM ZnCl_2 for 500 s to monitor dissociation. HRG concentrations in nM are indicated adjacent to each sensogram tracing. These are data from a single experiment, which was performed three times.

TABLE 1

Dissociation constants and stoichiometries for the binding of HRG to γ_A/γ_A -fibrinogen, γ_A/γ' -fibrinogen, or γ' -peptide

The binding of HRG to immobilized fibrinogen isoforms or γ' -peptide in the presence of 20 μM Zn^{2+} was quantified using SPR. K_d values were determined by kinetic analysis of the data, and stoichiometries were calculated according to the BIAtechnology handbook.

	K_d	Stoichiometry
	nM	HRG/molecule
γ_A/γ_A -Fibrinogen	8.8 ± 0.9	1.7 ± 0.3
γ_A/γ' -Fibrinogen	8.9 ± 3.9	3.2 ± 0.5
γ_A/γ_A -Fibrin	19.3 ± 2.6	1.8 ± 0.3
γ_A/γ' -Fibrin	10.9 ± 0.9	2.9 ± 0.6
γ' -Peptide	0.8 ± 0.01	0.7 ± 0.02

rected IgG did not alter the affinity of HRG for γ_A/γ' -fibrinogen. However, in the presence of the antibody, the molar stoichiometry of HRG for γ_A/γ' -fibrinogen was similar to that for γ_A/γ_A -fibrinogen (1.8 ± 0.5 and 1.8 ± 0.6 , respectively), providing further support for the concept that the γ' -chain on γ_A/γ' -fibrinogen represents a unique HRG binding site. These data suggest that HRG has multiple high-affinity binding sites on fibrinogen and that blocking one binding site has minimal effects on the others.

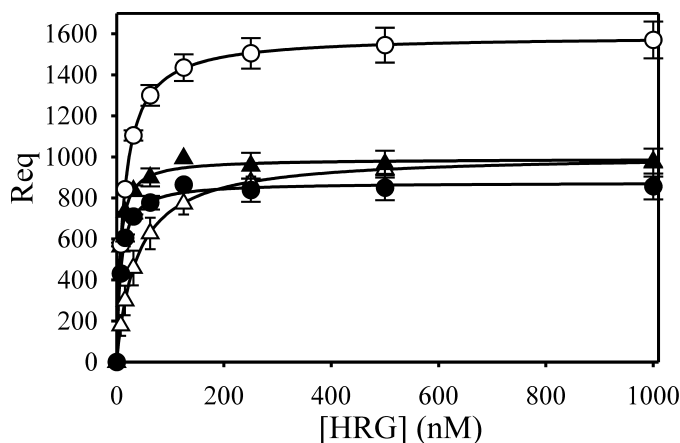


FIGURE 3. Effect of the γ' -peptide-directed IgG on HRG binding to γ_A/γ_A - or γ_A/γ' -fibrinogen. Flow cells containing immobilized γ_A/γ_A -fibrinogen (triangles) or γ_A/γ' -fibrinogen (circles) were pretreated with (closed) or without (open) 0.5 μM γ' -peptide-directed IgG before injection of HRG (0–1.0 μM). Symbols represent the mean $\pm$ S.D. of two experiments, and lines represent nonlinear regression analyses of the data.

HRG Binding to the γ' -Peptide—To confirm that the γ' -chain affords HRG an additional binding site, binding of HRG to synthetic γ' -peptide was examined. First, the interaction of HRG with fluorescein- γ' -peptide was examined by fluorescence. Neither HRG nor Zn^{2+} alone altered the fluorescence intensity of the f- γ' -peptide, suggesting that in the absence of Zn^{2+} , there is no interaction. However, when Zn^{2+} was titrated in the presence of HRG, the fluorescence intensity of fluorescein- γ' -peptide decreased in a dose-dependent and saturable manner (Fig. 4A), suggesting that Zn^{2+} facilitates the binding of HRG to the peptide. Nonlinear regression analysis of the data revealed that the apparent K_d for ZnCl_2 required to promote the HRG-fluorescein- γ' -peptide interaction was 9.1 ± 4.5 μM . This value is similar to the apparent K_d of 1.2 μM for Zn^{2+} -mediated promotion of HRG binding to γ_A/γ' -fibrinogen (Fig. 1).

To confirm these results, the interaction of HRG with γ' -peptide was examined by SPR. Biotinylated γ' -peptide was adsorbed to a streptavidin-modified flow cell, and HRG binding was monitored in the presence of 20 μM ZnCl_2 . HRG bound immobilized γ' -peptide in a concentration-dependent and saturable manner, and binding was blocked by the γ' -peptide-directed IgG (Fig. 4B). Based on kinetic analysis, HRG binds the γ' -peptide with a K_d value of 0.79 ± 0.01 nM in the presence of Zn^{2+} ; there is no detectable binding in the absence of Zn^{2+} (data not shown). These data offer independent confirmation that HRG binds to the COOH terminus of the γ' -chain of γ_A/γ' -fibrinogen in a Zn^{2+} -dependent fashion.

Effect of the γ' -Peptide-directed IgG on the Binding of HRG to γ_A/γ_A - or γ_A/γ' -Fibrin—Having shown that HRG binds fibrinogen with high affinity in a Zn^{2+} -dependent fashion, we next used SPR to determine the affinity of HRG for fibrin. To convert immobilized fibrinogen to fibrin, flow cells were treated with thrombin (30). HRG bound to both isoforms of fibrin with affinities (Table 1) and Req values similar to those for fibrinogen, suggesting that HRG binding is unaltered when fibrinogen is converted to fibrin. To complement the SPR studies, we also assessed the binding of ^{125}I -HRG to fibrin clots. Clots contain-

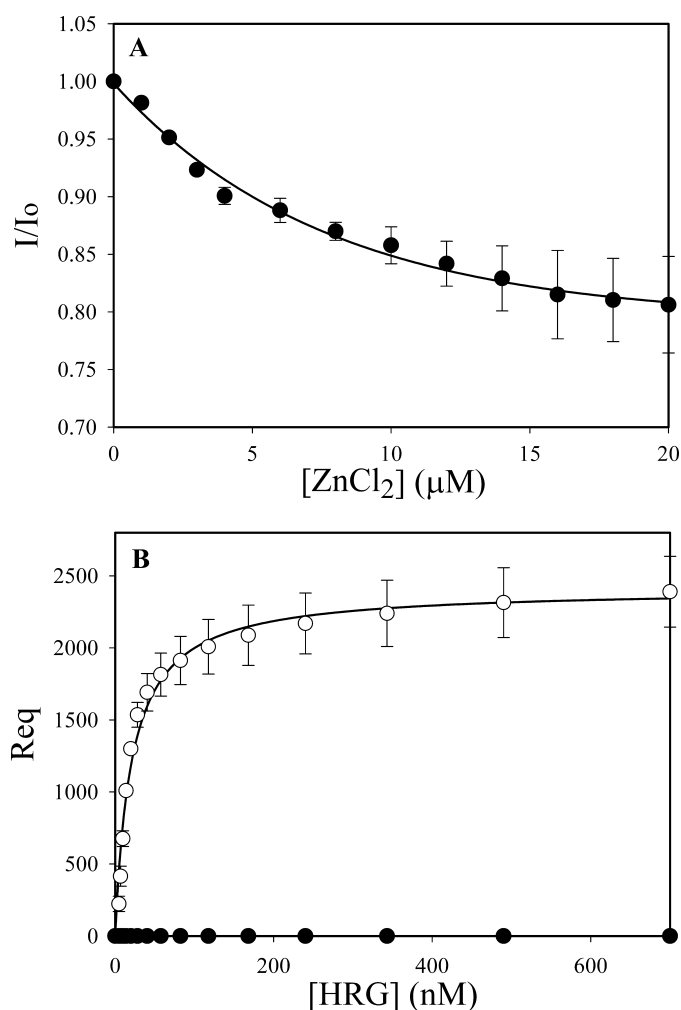


FIGURE 4. Binding of HRG to γ' -peptide. *A*, the binding of 1.1 μ M HRG to fluorescein-labeled γ' -peptide (50 nM) was monitored in the presence of ZnCl_2 (0–20 μ M) at $\lambda_{\text{ex}} = 492$ nm and $\lambda_{\text{em}} = 532$ nm. Initial fluorescence was determined in the presence of HRG but in the absence of ZnCl_2 (I_0). Aliquots of ZnCl_2 were then added, and the fluorescence intensity (I) was measured after each addition. I/I_0 values are plotted versus ZnCl_2 concentrations. *B*, binding of HRG to immobilized γ' -peptide was measured by SPR. Biotinylated γ' -peptide was adsorbed to a streptavidin-immobilized CM4 chip. Binding of HRG (0–700 nM) to γ' -peptide in the presence of 20 μ M ZnCl_2 was then determined in the absence (open symbols) or presence (closed symbols) of the γ' -peptide-directed IgG. Corrected Req values are plotted against the input HRG concentrations. Symbols represent the mean $\pm$ S.D. of two experiments, and lines represent nonlinear regression analyses of the data.

ing varying concentrations of fibrinogen were prepared, and the amount of ^{125}I -HRG in the supernatants of compacted clots was determined. In keeping with our SPR data, ^{125}I -HRG did not bind to fibrin clots in the absence of Zn^{2+} (data not shown). With 20 μ M Zn^{2+} , ^{125}I -HRG bound to γ_A/γ_A - and γ_A/γ' -fibrin clots with K_d values of 105.9 ± 21.0 and 33.3 ± 6.2 nM, respectively (Fig. 5A). Similar results were obtained in the reciprocal experiment using varying concentrations of ^{125}I -HRG and a fixed concentration of fibrinogen (data not shown). The binding constants obtained here are comparable with the previously reported K_d value of 250 nM for the interaction of ^{125}I -HRG with fibrin formed from unfractionated fibrinogen, which consists of both isoforms of fibrinogen (11). Taken together, these results suggest that HRG binds fibrinogen and remains bound when fibrinogen is converted to fibrin.

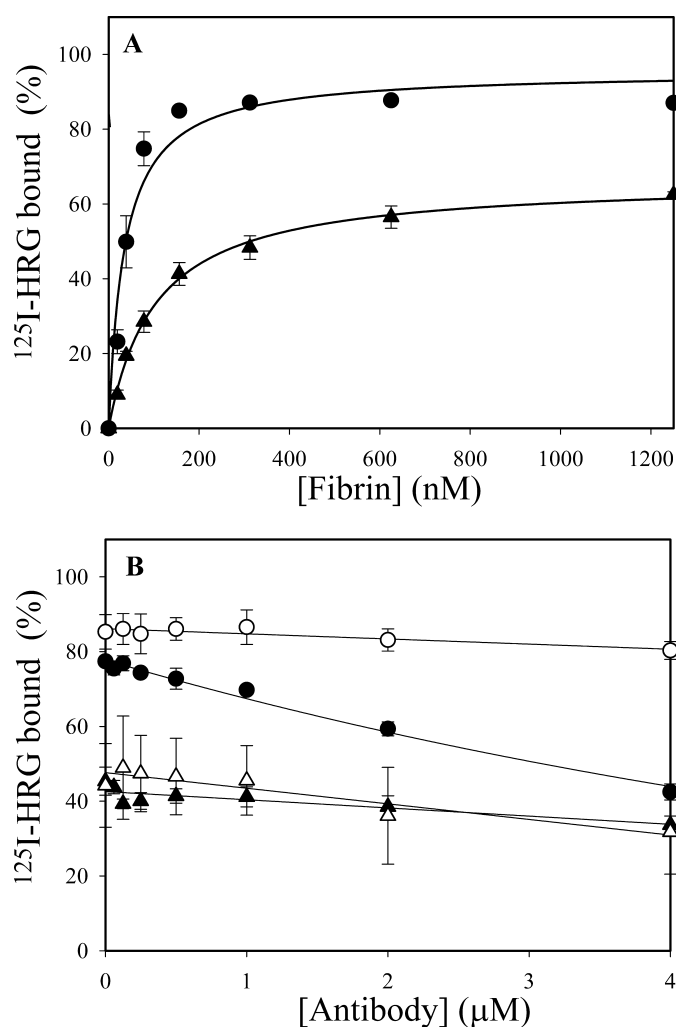


FIGURE 5. Binding of ^{125}I -HRG to γ_A/γ_A - or γ_A/γ' -fibrin clots. *A*, 0–1.25 μ M γ_A/γ_A - (triangles) or γ_A/γ' -fibrinogen (circles) was added to microcentrifuge tubes, and the binding of ^{125}I -HRG (40 nM) to clots was assessed after thrombin addition. *B*, the binding of ^{125}I -HRG (20 nM) to 0.25 μ M γ_A/γ_A - (triangles) or γ_A/γ' -fibrin clots (circles) was assessed in the presence of 0–4 μ M γ' -peptide-directed Fab fragments (closed symbols) or a control sheep IgG (open symbols). Experiments were performed in TBS-Tween containing 20 μ M ZnCl_2 and 2 mM CaCl_2 and clots were generated with 10 nM thrombin. After incubation at 23 $^\circ\text{C}$ for 45 min, fibrin clots were pelleted by centrifugation, and the amount of free ^{125}I -HRG in the supernatant was used to calculate the fraction bound. The percent of HRG bound to the clots is plotted versus fibrin or antibody concentrations. Symbols represent the mean $\pm$ S.D. of two experiments, each performed in duplicate, whereas the lines represent nonlinear regression analyses of the data.

Next, we examined the effect of varying concentrations of Fab fragments derived from the γ' -peptide-directed IgG on ^{125}I -HRG binding to γ_A/γ_A - or γ_A/γ' -fibrin clots. Fab fragments had minimal effects on HRG binding to γ_A/γ_A -fibrin clots (Fig. 5B). In contrast, at 4 μ M, the Fab fragments reduced HRG binding to γ_A/γ' -fibrin clots by 50%, providing further evidence that HRG binds to the γ' -chain of γ_A/γ' -fibrin. A non-immune sheep IgG was used as a control and demonstrated no effect.

HRG Binding to Fibrinogen Fragments—To localize the HRG binding domains on fibrinogen, binding of HRG to immobilized fibrinogen fragments was examined by SPR (data not shown). To avoid potential contribution of the γ' -chain to HRG binding, fragments X, D, and E were prepared from γ_A/γ_A -

HRG Modulates Thrombin Binding to Fibrin

fibrinogen. In the presence of Zn^{2+} , HRG bound fragment X with a K_d value of 63.5 ± 11.8 nM, suggesting that the αC -domain of fibrinogen does not represent the primary HRG binding site. HRG also bound fragments D and E with high affinity in a Zn^{2+} -dependent manner, with K_d values of 8.0 ± 1.3 and 23.3 ± 2.2 nM, respectively. As a negative control, we demonstrated that HRG did not bind immobilized FPRck-thrombin in the absence or presence of Zn^{2+} . As a positive control, we showed that FPRck-thrombin bound fragment E, with a K_d value of 5.0 ± 0.4 μM , but did not bind fragment D, findings in agreement with previously published results (25). Therefore, these data suggest that, in addition to its interaction with the γ' -chain, HRG binds to other unique sites on fibrinogen.

Diffusion of HRG from Fibrin Clots—To identify differences in the binding of HRG to γ_A/γ_A - and γ_A/γ' -fibrin clots, we monitored the dissociation of ^{125}I -HRG from preformed fibrin clots (Fig. 6). Diffusion in the presence of 2 M NaCl and 2 mM EDTA served as the base-line control because ionic and divalent cation-dependent interactions are abrogated (10, 30). In the presence of Zn^{2+} , the rates of diffusion of ^{125}I -HRG from γ_A/γ_A - and γ_A/γ' -fibrin clots were significantly ($p < 0.05$) slowed by 3- and 11-fold, respectively, compared with those determined in the presence of diethyldithiocarbamate trihydrate, a specific Zn^{2+} chelator (32). Consistent with the concept that γ_A/γ' -fibrin affords HRG an additional binding site, the rate of ^{125}I -HRG diffusion from γ_A/γ' -fibrin was 3-fold slower than that from γ_A/γ_A -fibrin ($p < 0.05$). Collectively, our results offer independent confirmation that the HRG-fibrinogen interaction is Zn^{2+} -dependent and that there is an additional HRG binding site on γ_A/γ' -fibrin.

Effect of FPRck-thrombin on HRG Binding to the γ' -Chain—In addition to binding HRG, the γ' -chain COOH extension binds thrombin (5, 9). SPR was used to determine whether the two proteins compete for binding to this region. This approach exploits the fact that the dissociation rate of HRG from immobilized γ' -peptide is much slower than that of thrombin. HRG (1 μM) was injected into flow cells containing immobilized biotinylated γ' -peptide, and the subsequent dissociation phase was monitored in the absence or presence of FPRck-thrombin or prothrombin in concentrations up to 8.0 μM . Prothrombin was used as a negative control because it does not bind to the γ' -peptide (33). Whereas FPRck-thrombin displaced HRG from the γ' -peptide in a concentration-dependent manner, prothrombin did not (Fig. 7). These data confirm that thrombin and HRG compete for binding to the γ' -peptide.

To determine whether the same was true with fibrin clots, we next examined the effect of increasing concentrations of FPRck-thrombin on ^{125}I -HRG binding to γ_A/γ_A - or γ_A/γ' -fibrin clots. At 10 μM , FPRck-thrombin reduced the amount of HRG bound to γ_A/γ' -fibrin clots by 90% but only reduced HRG bound to γ_A/γ_A -fibrin clots by 15% (Fig. 8A). Similar results were obtained in the reciprocal competition experiments using varying concentrations of HRG and a fixed concentration of ^{125}I -YPRck-thrombin (Fig. 8B). Collectively, these data confirm that HRG and thrombin compete for binding to the γ' -chain on γ_A/γ' -fibrin(ogen) in a mutually exclusive fashion.

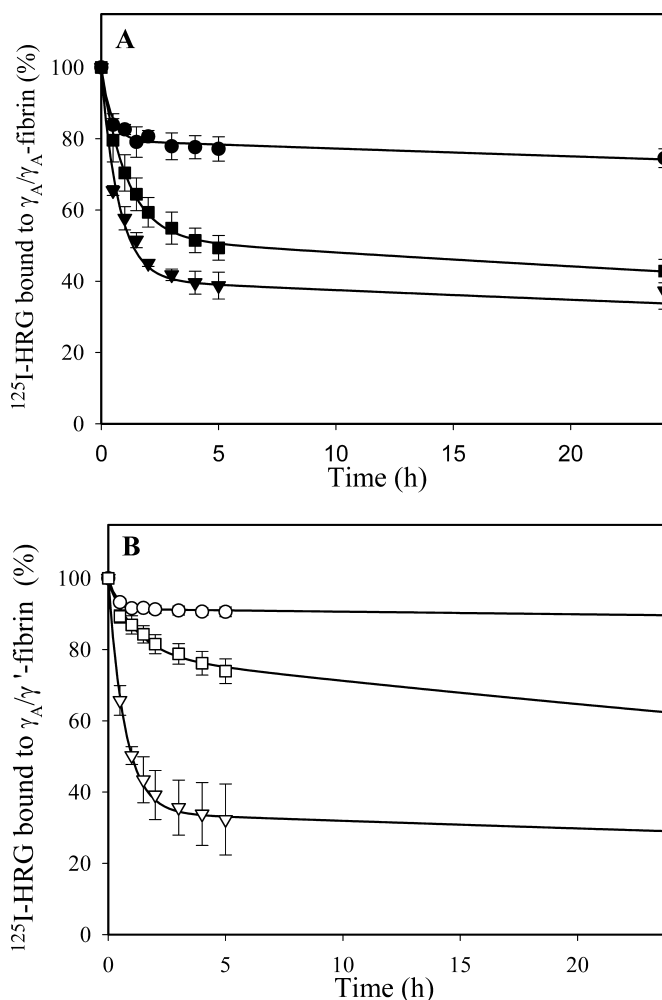


FIGURE 6. Dissociation of ^{125}I -HRG from preformed γ_A/γ_A - or γ_A/γ' -fibrin clots. Aliquots of 5 μM γ_A/γ_A - (A) or γ_A/γ' -fibrinogen (B) containing 2 mM CaCl_2 , 20 μM ZnCl_2 , 20 nM factor XIII, and 50 nM ^{125}I -HRG were clotted with 10 nM thrombin around plastic inoculation loops. After incubation at 23 $^\circ\text{C}$ for 45 min, clots were counted for radioactivity and incubated in tubes containing 5 ml of 2 M NaCl and 2 mM of EDTA (triangles), 2 mM CaCl_2 and 10 μM diethyldithiocarbamate trihydrate (squares), or 2 mM CaCl_2 and 20 μM ZnCl_2 (circles). At intervals, clots were removed, and residual radioactivity was used to determine the percent of ^{125}I -HRG that remained clot-associated. The symbols represent the mean $\pm$ S.D. of three experiments, whereas the lines represent nonlinear regression analyses of the data using two-component exponential decay model.

Detection of HRG-Fibrinogen Complexes in Plasma—Because HRG binds fibrinogen with high affinity, it was of interest to determine whether HRG-fibrinogen complexes can be detected in plasma by immunoassay. Plasma was first dialyzed to remove citrate and then reconstituted with 18 μM ZnCl_2 . The concentration of HRG-fibrinogen complexes detected in normal plasma was 1 μM , whereas no complexes were detected in HRG-deficient plasma. Because the plasma concentration of HRG ranges from 1.6 to 2 μM (12, 13, 24), our findings suggest that in the presence of Zn^{2+} ~50–60% of HRG in plasma circulates in complex with fibrinogen.

DISCUSSION

Despite increasing evidence that the interaction of HRG with fibrin(ogen) plays an important role in innate immunity, inflammation, and coagulation, little is known about this inter-

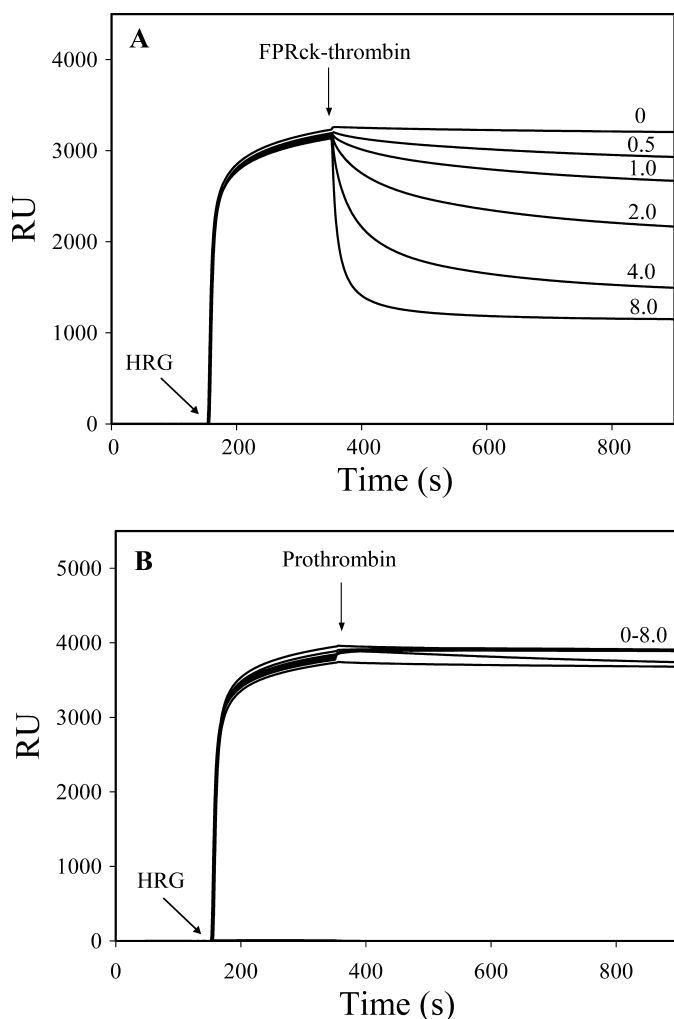


FIGURE 7. Effect of FPRck-thrombin or prothrombin on the interaction of HRG with γ' -peptide. Biotinylated γ' -peptide was immobilized on a streptavidin-modified flow cell to 200 RU, and an unmodified flow cell served as a control. Arrows indicate injection of 1 μ M HRG for 200 s followed by injections of FPRck-thrombin (panel A) or prothrombin (panel B) at the concentrations (μ M) indicated. Injections were carried out in the presence of 20 μ M Zn^{2+} . These are data from a single experiment, which was performed three times.

action. To address this gap, we characterized the binding of HRG to γ_A/γ_A - and γ_A/γ' -forms of fibrinogen and fibrin. In the presence of physiological concentrations of Zn^{2+} , HRG binds γ_A/γ_A - and γ_A/γ' -fibrinogen with similar affinities (K_d values of ~ 9 nM). The affinities of HRG for both isoforms of fibrin are comparable to those for fibrinogen, suggesting that the conversion of fibrinogen to fibrin does not alter the binding of HRG. In addition to its interaction with the unique COOH terminus of the γ' -chain of γ_A/γ' -fibrin(ogen), HRG binds to fragments D and E, suggesting that there are several HRG binding sites on fibrinogen. In support of these observations, HRG-fibrinogen complexes were detected in plasma. Taken together, our findings suggest that HRG circulates in plasma bound to fibrinogen and that the complex remains intact when fibrinogen is converted to fibrin.

The absolute requirement for Zn^{2+} to promote the HRG-fibrinogen interaction underscores the regulatory role this cation may have in hemostasis. Zn^{2+} is important for the binding of HRG to bacteria, cells, and hemostatic factors, such as glyco-

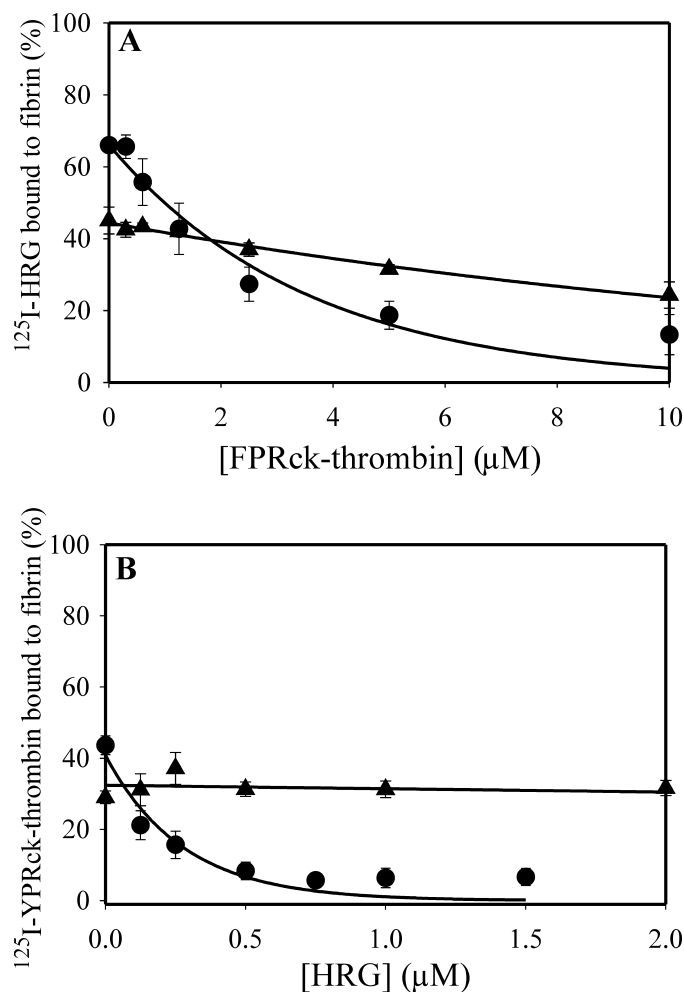


FIGURE 8. Effect of competitors on the binding of ^{125}I -HRG or ^{125}I -YPRck-thrombin to γ_A/γ_A - or γ_A/γ' -fibrin clots. A, ^{125}I -HRG (20 nM) was added to microcentrifuge tubes containing 2 μ M γ_A/γ_A -fibrinogen (triangles) or γ_A/γ' -fibrinogen (circles) in the presence of FPRck-thrombin (0–10 μ M). B, ^{125}I -YPRck-thrombin (20 nM) was added to microcentrifuge tubes containing 2 μ M γ_A/γ_A -fibrinogen (triangles) or 0.25 μ M γ_A/γ' -fibrinogen (circles) in the presence of HRG (0–2 μ M). In both experiments 2 mM CaCl_2 plus 20 μ M ZnCl_2 were present and clotting was initiated with 10 nM thrombin. After incubation at 23 $^\circ\text{C}$ for 45 min, fibrin was pelleted by centrifugation, and free ^{125}I -HRG or ^{125}I -YPRck-thrombin in the supernatant was used to calculate the bound fraction. The percent of fibrin-bound ^{125}I -HRG or ^{125}I -YPRck-thrombin is plotted versus the FPRck-thrombin or HRG concentration, respectively. The symbols represent the mean $\pm$ S.D. of two experiments, each performed in duplicate, whereas the lines represent nonlinear regression analyses of the data.

saminoglycans, plasminogen, and factor XIIa (12, 19, 24, 34). It is likely that there are still other HRG interactions that have been overlooked because of the widespread use of citrate as an anticoagulant. The total Zn^{2+} concentration in plasma is ~ 20 μ M, and the majority of Zn^{2+} is bound to albumin (35). Although the concentration of Zn^{2+} that is not bound to proteins is only 0.5–1 μ M (31, 36), the free Zn^{2+} concentration can increase under a variety of conditions. For example, platelets can secrete Zn^{2+} when they are activated at sites of vascular injury (37). Furthermore, when fatty acids bind to albumin, they displace Zn^{2+} from the protein, thereby providing another mechanism whereby the concentration of free Zn^{2+} in plasma can be augmented (35). In addition to alterations in the Zn^{2+} concentration, the activity of HRG can also be modulated by changes in pH, with optimal binding to ligands observed under

more acidic conditions. Consequently, the decrease in pH that occurs with reduced tissue perfusion may also enhance the affinity of HRG for both Zn^{2+} and its ligands (14). Therefore, the current data extend the concept that Zn^{2+} serves as a dynamic switch that regulates HRG activity and directs it to various pathways involved in hemostasis (14, 34).

Although we show that Zn^{2+} is essential for the interaction of HRG with fibrin(ogen), the mechanism by which Zn^{2+} mediates this binding is unknown. Both HRG and fibrin(ogen) bind Zn^{2+} (37, 38). Therefore, Zn^{2+} may act as a cofactor that simultaneously binds HRG and fibrin(ogen) in a coordinated fashion (39). Alternatively, Zn^{2+} binding to HRG may induce conformational changes that facilitate its interaction with fibrin(ogen), a concept supported by the observation that Zn^{2+} alters the conformation of a synthetic His-Pro-rich peptide (16). Furthermore, the intrinsic fluorescence of HRG decreases upon Zn^{2+} titration (data not shown), providing additional support for the notion that Zn^{2+} alters HRG conformation. Binding of Zn^{2+} to the HRR domain of HRG indirectly promotes the interaction of heparan sulfate or plasminogen with the NH_2 -terminal cystatin domains of HRG, suggesting that Zn^{2+} binding to the HRR domain modulates other domains (15, 40). These data point to a mechanism whereby Zn^{2+} modulates the structure and function of HRG.

Because HRG binds to the γ' -chain on γ_A/γ' -fibrin(ogen), novel roles of HRG can be envisioned. In addition to binding HRG, the COOH terminus of the γ' -chain also binds thrombin and factor XIIIa and, by so doing, may modulate coagulation and fibrinolysis (10, 41). The importance of fibrin as a reservoir of thrombin is highlighted by the observation that thrombi harvested at autopsy contain abundant amounts of active thrombin (42). Fibrin-bound thrombin has been postulated to be an important mediator of thrombus expansion because of its capacity to locally activate platelets and to promote its own generation through activation of factor V and factor VIII (43, 44). The procoagulant activity of thrombin bound to γ_A/γ' -fibrin appears to be greater than that of thrombin bound to γ_A/γ_A -fibrin because the γ' -chain mediates the high affinity interaction of thrombin with fibrin, and γ_A/γ' -fibrin affords bound thrombin more protection from inhibition by the anti-thrombin-heparin complex than γ_A/γ_A -fibrin (10). Further support for this concept comes from epidemiological studies that suggest that higher circulating levels of γ_A/γ' -fibrinogen are associated with an increased risk of cardiovascular disease (45, 46). HRG competes with thrombin for binding to the γ' -chain as evidenced by its capacity to displace FPRck-thrombin from γ' -peptide or from γ_A/γ' -fibrin clots. Consequently, by displacing fibrin-bound thrombin, HRG may have anti-thrombotic properties. HRG may also compete with factor XIII, which is proposed to bind the γ' -chain (47), thereby attenuating fibrin cross-linking and endowing HRG with pro-fibrinolytic activity. Studies in HRG-deficient mice support the concept that HRG affects hemostasis. HRG-deficient mice have a shorter prothrombin time and a longer bleeding time than their wild-type counterparts. In addition, thrombi formed in HRG-deficient mice are more susceptible to fibrinolysis than those generated in control mice (17). The contribution of the HRG

interaction with the γ' -chain to the anticoagulant and anti-fibrinolytic activities of HRG remains to be determined.

In addition to its role in hemostasis, fibrinogen appears to be an important mediator of inflammation and innate immunity because fibrinogen and fibrin stimulate peripheral blood mononuclear cells and vascular smooth muscle cells to synthesize proinflammatory cytokines (48–50). Furthermore, bacteria trapped within fibrin clots are protected from host defenses and the action of antibiotics (51). However, because HRG has antimicrobial properties, the HRG-fibrin interaction promotes bacterial entrapment and killing (21). The capacity of HRG to enhance bacterial killing has been localized to its NH_2 -terminal and HRR domains. Thus, in the presence of Zn^{2+} or when the pH is low, HRG induces lysis of the bacterial cell wall (19, 20). These conditions can occur at sites of injury or wound-healing where activated platelets release Zn^{2+} and local ischemia lowers pH (13, 52). Activated platelets also release HRG (13) and, by so doing, may amplify the antimicrobial effect. Our observation that HRG binds fibrinogen and fibrin provides a regulatory mechanism by which HRG may mediate bacterial killing within a clot. The importance of HRG in modulating the inflammatory response has been confirmed in mouse models. Thus, compared with wild-type mice, HRG-deficient mice given subcutaneous injections of *S. pyogenes* exhibit attenuated abscess formation and reduced recruitment of neutrophils and macrophages to the site of infection (21), suggesting that HRG modulates the inflammatory response. In support of this concept, a synthetic peptide analog of the HRR of HRG attenuated the secretion of interleukin-8 from lipopolysaccharide-stimulated, CD14-transfected monocytes (53). These observations raise the possibility that HRG plays a part in the inflammatory response to infection.

Although HRG is an abundant plasma protein with multiple ligands, its physiological role remains unknown (for review, see Ref. 18). HRG is hypothesized to be an important effector of hemostasis and immunity. The ability of HRG to bind fibrinogen and displace thrombin from fibrin may provide an important link between these two systems and reveals a potential mechanism by which this could occur. Additional regulation may result from variations in the local pH and/or Zn^{2+} concentration.

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